Round-14 Delta after the G2a Failure: Withdrawing Literal D2 and Restating Coverage by Relation

Repair note following targeted G2/G3 review

July 8, 2026

This is an emergency delta, not a completed round-14 proof. It records the counterexample to round 13's extraction theorem and the smallest repair shape I currently see. The status of the decidability argument remains strongly supported, not certified.

1 Retraction of the round-13 D2 extraction claim

Round 13's D2 clause said that whenever an occurrence x has nonempty obligation set, a shadow of x is propagated in every direction out of every bag where x is live or shadowed; if a crossing would require a non-horizontal row entry, the crossing is replaced by verticalization, inserting x as a port in the target bag.

This literal clause is not compatible with finite-width extraction. The fixed satisfiable concept

$$C_{\mathrm{G2a}} = \exists \mathsf{PP}.\top \ \sqcap\ \forall \mathsf{PP}.\big(\exists \mathsf{PP}.\top \ \sqcap\ (\forall \mathsf{PP}.\exists \mathsf{PP}.\top \ \sqcap\ \forall \mathsf{DR}.A)\big)$$

forces an infinite chain $a_0 \mathsf{PP} a_1 \mathsf{PP} a_2 \mathsf{PP} \cdots$, and every a_i for $i \geq 1$ carries the horizontal universal $\forall \mathsf{DR}.A$. In the edge-bag presentation of the length-m prefix, source a_k crossing downward over separator a_i has relation $a_k \mathsf{PPI} a_i$, hence a D1-forbidden vertical row entry. Literal D2 therefore inserts a_k into every lower edge bag, including the root edge bag. The root edge bag has size $m+1$ for arbitrary m.

The script `verification/wp18_g2a_verticalization_width_counter.py` checks the oracle satisfiability and the finite-prefix width growth. Therefore round 13's Theorem B(a), D-form, and Lemma 6.1 are withdrawn as stated.

2 Principle of the repair

The mistake was to make coverage occurrence-wide. The truth lemma needs coverage only for a particular universal formula $\forall R.D$ and a target y satisfying $\rho_T(x, y) = R$. Pairs whose relation is not R do not need to be represented for that obligation.

Round 14 should split obligations by relation:

$$\Omega^H_x = \{(R, D) \in \Omega_x : R \in \{\mathsf{DR}, \mathsf{PO}\}\}, \qquad \Omega^V_x = \{(R, D) \in \Omega_x : R \in \{\mathsf{PP}, \mathsf{PPI}\}\}.$$

Vertical obligations are handled by type propagation along uniformly oriented vertical support chains. Horizontal obligations are handled by horizontal rows only at open horizontal frontiers. A vertical region that is certified to be entirely below or above the source is a barrier for the horizontal obligation, not a reason to insert the remote source as a live port.

3 Replacement for D2: draft D2H

The following clause is the proposed replacement. It is deliberately stated as a proof obligation, not as a proved theorem.

(D2H) Horizontal-frontier propagation with vertical barriers. For every source occurrence x and every horizontal obligation $(R, D) \in \Omega_x^H$, the abstraction carries a shadow state for x across all adjacent components that are not certified vertical barriers for x. At a crossed bag, the declared row entries are exactly the horizontal true relations from x to the current ports relevant to that frontier, and D1/D5 apply to these entries.

If a boundary crossing would require a vertical entry from x to every continued separator port, the transition may take a vertical-barrier outcome instead of inserting x as a port. The barrier record stores only a finite certificate: orientation ($x\mathsf{PP}$-above or $x\mathsf{PPI}$-above the separator), the separator profile, and the obligation kind being protected. It is not part of Λ and is not a shadow row. If a later side context can contain a DR or PO target for x, the saturation certificate must reopen a horizontal frontier and resume row propagation there.

This clause is meant to avoid the G2a tower: each ancestor a_k sees the lower chain as a vertical barrier, so it does not get inserted into the root bag merely because it carries $\forall\mathsf{DR}.A$.

4 Replacement for D3: explicit vertical coverage

Round 13's D3 was sound but underspecified as a coverage device. The amended version should be:

(D3V) Uniform vertical propagation. If $\forall\mathsf{PP}.D \in \mathrm{tp}(p)$ and a co-bagged vertical support edge has $N(p, q) = \mathsf{PP}$, then $D, \forall\mathsf{PP}.D \in \mathrm{tp}(q)$; dually for PPI. The valid abstraction must certify that every target relevant to a vertical universal is co-bagged with the source or lies on a uniformly oriented chain of such support edges. Mixed PP/PPI paths do not count as vertical coverage.

The last sentence is necessary. The probe `verification/wp19_g3_mixed_orientation_probe.py` checks a closed network with $x\mathsf{PP}z$, $z\mathsf{PPI}y$, and $x\mathsf{PP}y$; D3 propagates $\forall\mathsf{PP}.D$ from x to z but not to y.

5 Replacement Coverage Lemma

The old occurrence-wide lemma should be replaced by the following relation-indexed statement.

For every occurrence x, every formula $\forall R.D \in \mathrm{tp}(x)$, and every occurrence y with $\rho_T(x, y) = R$, at least one of the following holds:

(a) x and y are co-bagged, and the in-bag universal test forces $D \in \mathrm{tp}(y)$;

(b) $R \in \{\mathsf{DR}, \mathsf{PO}\}$ and the pair is covered by a declared horizontal row whose value is realized by Shadow Amalgamation;

(c) $R \in \{\mathsf{PP}, \mathsf{PPI}\}$ and y lies on a certified co-bagged uniformly oriented vertical support chain from x, so D3V forces $D \in \mathrm{tp}(y)$.

Pairs outside the queried relation R need not be co-bagged or row-covered. This is the essential change that prevents the G2a tower from being forced into one bag.

6 New proof obligations

The repair is viable only if the following obligations are discharged.

1. **Barrier soundness.** A vertical-barrier certificate must imply that no target in the skipped component can be in the queried horizontal relation to x, unless a later certified side exit reopens a horizontal frontier.

2. **Barrier completeness.** In every saturated split-forest presentation extracted from a model, every horizontal target of every horizontal universal is either co-bagged or reached by D2H row propagation.

3. **Width.** Barrier records consume finite state but no port manifestations. Port insertions must again be chargeable to the witness-generated reasons (g1)–(g5); remote sources crossing a vertical tower must not be charged to the receiving bag.

4. **Automaton locality.** D2H barriers and side exits must be checkable by finite two-way states over bounded bag labels.

5. **Truth lemma.** The universal case must use the relation-indexed Coverage Lemma above, not the old occurrence-wide lemma.

7 Status

This delta identifies the failed round-13 clause and a plausible direction for round 14. It does not certify that the barrier mechanism is complete. The Shadow Amalgamation Lemma for horizontal rows remains available as a component; the remaining load is the new extraction and relation-indexed coverage proof.